

# Contoso Electronics

Equipment Reference & Maintenance Guide

## Digital Storage Oscilloscope

Contoso WaveView DSO-4000X

### Asset Identification

| Field             | Detail                                         |
|-------------------|------------------------------------------------|
| Asset ID          | CE-OSC-1200                                    |
| Category          | Test & Measurement                             |
| Manufacturer      | Contoso Instruments Division                   |
| Location          | Electrical Test Lab - Bench 3                  |
| Installation Date | 2023-03-14                                     |
| Warranty Expiry   | 2027-03-14                                     |
| Support Contact   | test.equipment.support@contoso.com   ext. 4210 |

### Overview

The Contoso WaveView DSO-4000X is a 4-channel digital storage oscilloscope used for capturing, visualizing, and analyzing electrical signals during board bring-up, quality assurance, and failure analysis. It supports advanced triggering, protocol decoding, and automated measurements for high-speed digital and analog designs.

### Technical Specifications

| Parameter           | Value                          |
|---------------------|--------------------------------|
| Bandwidth           | 1 GHz                          |
| Channels            | 4 analog + 16 digital (MSO)    |
| Sample Rate         | 5 GSa/s per channel            |
| Memory Depth        | 50 Mpts                        |
| Vertical Resolution | 8-bit (12-bit high-res mode)   |
| Display             | 12.1 in capacitive touchscreen |
| Input Impedance     | 1 MΩ / 50 Ω selectable         |
| Power               | 100-240 VAC, 50/60 Hz, 200 W   |

### Preventive Maintenance Schedule

| Interval | Task                                                     |
|----------|----------------------------------------------------------|
| Daily    | Wipe screen and inspect probes for damage.               |
| Monthly  | Verify self-calibration routine completes without error. |

|           |                                                           |
|-----------|-----------------------------------------------------------|
| Quarterly | Clean cooling fan intake and check firmware version.      |
| Annual    | Full metrology calibration by accredited lab (traceable). |

Safety Guidelines

- Never exceed the maximum rated input voltage on any channel.
- Use only properly rated probes and attenuators.
- Ensure the instrument is properly grounded before use.
- Do not operate with the enclosure removed.

Common Issues & Troubleshooting

| Symptom                  | Recommended Action                                                         |
|--------------------------|----------------------------------------------------------------------------|
| No trace on screen       | Check trigger settings and channel enable; verify probe attenuation ratio. |
| Self-cal fails           | Allow 20-minute warm-up, disconnect all inputs, retry calibration.         |
| Distorted waveform       | Inspect probe grounding and compensation; replace worn probe tips.         |
| Touchscreen unresponsive | Reboot instrument; if persistent, log a work order for panel service.      |

Contoso Electronics internal maintenance documentation. For service requests, contact [test.equipment.support@contoso.com](mailto:test.equipment.support@contoso.com) | ext. 4210 or submit a work order through the maintenance portal.